FINAL PROJECT ASSIGNMENT

End-to-End Data Analysis Workflow

Project Overview

You will complete an **end-to-end data analysis project** using a dataset of your choice. The project must demonstrate all the skills you've learned in this course:

Prepare → Analyze → Visualize → Predict → Evaluate Responsibly

Machine Learning is OPTIONAL - you can complete the entire project without ML.

Timeline

Milestone	Date	Weight
Milestone 0: Dataset Selection & Quality Audit	Week 3	5%
Milestone 1: Cleaned Data & Statistics	Week 6	10%
Milestone 2: Static Visualization Story	Week 10	15%
Milestone 3: Interactive Dashboard Prototype	Week 12	15%
Final Project: Complete Report & Presentation	Week 16	25%
Total		**70%**

WHAT YOU ACTUALLY NEED TO DO

Step 1: Choose a Dataset (Milestone 0)

Task: Select a dataset and document it.

What to submit:

1. **Dataset name and source** (provide URL)
2. **Data dictionary**: List all columns with descriptions
3. **Initial quality audit**:
    - How many rows and columns?
    - What data types?
    - How many missing values?
    - Any obvious problems?

Example submission:

```
# Student submission for Milestone 0

"""
DATASET: Titanic Passenger Data
SOURCE: https://www.kaggle.com/c/titanic
DESCRIPTION: Passenger manifest from Titanic, including survival status

DATA DICTIONARY:
- PassengerId: Unique ID for each passenger
- Survived: 0 = No, 1 = Yes
- Pclass: 1st, 2nd, 3rd class
- Name: Passenger name
- Sex: male/female
- Age: Age in years
- SibSp: Number of siblings/spouses aboard
- Parch: Number of parents/children aboard
- Ticket: Ticket number
- Fare: Ticket fare
```

```
- Cabin: Cabin number
- Embarked: C = Cherbourg, Q = Queenstown, S = Southampton

INITIAL QUALITY AUDIT:
- Rows: 891
- Columns: 12
- Missing values:
  - Age: 177 missing (19.9%)
  - Cabin: 687 missing (77.1%)
  - Embarked: 2 missing (0.2%)
- Data types: 5 numeric, 7 object
- Duplicates: 0
- Outliers: Fare has extreme values (max = 512.33)
"""
```

Step 2: Clean the Data (Milestone 1)

Task: Clean the dataset and perform statistical analysis.

What to submit:

1. **Cleaned dataset** (CSV file)
2. **Documented cleaning decisions** - What did you change and why?
3. **Descriptive statistics** - Mean, median, min, max, etc.
4. **Inferential analysis** - At least one statistical test (t-test, ANOVA, chi-square, etc.)

Example submission:

```
# Student submission for Milestone 1

# 1. Cleaning decisions
"""
CLEANING DECISIONS:
1. Dropped 'Cabin' column (77% missing — too many to impute)
2. Imputed missing 'Age' using median age by passenger class
   - 1st class median: 37
   - 2nd class median: 29
```

```
   - 3rd class median: 24
3. Imputed missing 'Embarked' with mode ('S')
4. Removed no duplicate rows
5. Identified fare outliers (>$300) but kept them as legitimate data
"""

# 2. Load and clean
import pandas as pd
import numpy as np

df = pd.read_csv('titanic.csv')

# Clean data
df_clean = df.drop(columns=['Cabin'])
df_clean['Age'] = df_clean.groupby('Pclass')['Age'].transform(
    lambda x: x.fillna(x.median())
)
df_clean['Embarked'] = df_clean['Embarked'].fillna('S')

# 3. Descriptive statistics
print("Descriptive Statistics:")
print(df_clean[['Age', 'Fare', 'SibSp', 'Parch']].describe())

# 4. Inferential analysis (t-test)
from scipy import stats

# Compare age of survivors vs non-survivors
survivors = df_clean[df_clean['Survived'] == 1]['Age']
non_survivors = df_clean[df_clean['Survived'] == 0]['Age']
t_stat, p_value = stats.ttest_ind(survivors, non_survivors)

print(f"\nT-test results:")
print(f"Survivors average age: {survivors.mean():.1f}")
print(f"Non-survivors average age: {non_survivors.mean():.1f}")
print(f"p-value: {p_value:.4f}")
print(f"Conclusion: {'Significant' if p_value < 0.05 else 'Not significant'}
difference")
```

What you need to know:

- You must justify every cleaning decision
- You must include at least one statistical test
- You must interpret the results

Step 3: Create Static Visualizations (Milestone 2)

Task: Create a set of static visualizations telling a story about the data.

What to submit:

1. **At least 5 static visualizations** (as PNG/PDF files)
2. **Design rationale** - Why did you choose each chart type?
3. **Accessibility check** - Are your charts colorblind-friendly?
4. **Bias critique** - What might be misleading about your charts?

Example submission:

```
# Student submission for Milestone 2

import matplotlib.pyplot as plt
import seaborn as sns

# Set colorblind-friendly palette
colors = sns.color_palette("colorblind")

# Chart 1: Survival by Class
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Survival by passenger class
df_clean.groupby('Pclass')['Survived'].mean().plot(
    kind='bar', ax=axes[0,0], color=colors[0]
)
axes[0,0].set_title('Survival Rate by Passenger Class')
axes[0,0].set_xlabel('Passenger Class')
axes[0,0].set_ylabel('Survival Rate')
axes[0,0].set_ylim(0, 1)
```

```
# 2. Age distribution by survival
df_clean[df_clean['Survived'] == 1]['Age'].hist(
    alpha=0.5, label='Survived', bins=20, ax=axes[0,1], color=colors[1]
)
df_clean[df_clean['Survived'] == 0]['Age'].hist(
    alpha=0.5, label='Did Not Survive', bins=20, ax=axes[0,1], color=colors[2
]
)
axes[0,1].set_title('Age Distribution by Survival')
axes[0,1].set_xlabel('Age')
axes[0,1].set_ylabel('Count')
axes[0,1].legend()

# 3. Fare by class
df_clean.boxplot(column='Fare', by='Pclass', ax=axes[1,0])
axes[1,0].set_title('Fare Distribution by Class')
axes[1,0].set_xlabel('Passenger Class')
axes[1,0].set_ylabel('Fare ($)')

# 4. Survival by sex and class
survival_by_sex_class = df_clean.groupby(['Sex', 'Pclass'])['Survived'].mean(
).unstack()
survival_by_sex_class.plot(kind='bar', ax=axes[1,1], color=colors[3:6])
axes[1,1].set_title('Survival Rate by Sex and Class')
axes[1,1].set_xlabel('Sex')
axes[1,1].set_ylabel('Survival Rate')
axes[1,1].legend(title='Class')
axes[1,1].set_ylim(0, 1)

plt.tight_layout()
plt.savefig('milestone2_visualizations.png', dpi=300)

"""
DESIGN RATIONALE:
1. Bar chart for class survival - easy to compare categories
2. Histogram for age - shows distribution patterns
3. Box plot for fare - shows spread and outliers
4. Grouped bar chart - shows interaction between two variables
```

```
ACCESSIBILITY NOTES:
- Used colorblind-friendly palette (ColorBrewer)
- Added clear labels and titles
- Used patterns where possible
- High contrast between colors

BIAS CRITIQUE:
- Bar charts can exaggerate small differences
- Box plots hide individual data points
- Histogram bin size can affect interpretation
- Should include confidence intervals
"""
```

Step 4: Create Interactive Dashboard (Milestone 3)

Task: Create an interactive visualization/dashboard.

What to submit:

1. **Working interactive prototype** (HTML file or working code)
2. **Peer usability feedback** and your revision plan
3. **Explanation** of what interaction adds to the analysis

Example submission:

```
# Student submission for Milestone 3

import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# Interactive scatter plot
fig = px.scatter(
    df_clean,
    x='Age',
    y='Fare',
    color='Survived',
```

```
    size='Pclass',
    hover_data=['Name', 'Sex', 'Embarked'],
    title='Interactive: Age vs Fare by Survival Status',
    labels={'Survived': 'Survived?'},
    color_discrete_map={0: 'red', 1: 'green'}
)

fig.update_layout(
    height=600,
    width=800,
    title_font_size=20
)

# Save as HTML
fig.write_html('interactive_dashboard.html')

print("Interactive dashboard saved as 'interactive_dashboard.html'")

"""
INTERACTION FEATURES:
- Hover to see passenger details
- Zoom and pan to explore
- Color coding for survival
- Size encoding for passenger class

WHAT INTERACTION ADDS:
- Explore individual data points
- Discover patterns not visible in static plots
- Filter and subset data dynamically
- Share findings interactively

PEER FEEDBACK:
1. "Add filtering by class" - Implemented dropdown filter
2. "Make colors more intuitive" - Changed to red/green
3. "Show more data on hover" - Added name and embarked

REVISION PLAN:
- Add dropdown for filtering by sex
- Add option to change chart type
```

```
- Improve mobile responsiveness
"""
```

Step 5: Final Project Submission (Week 16)

Task: Submit complete project with report and presentation.

What to submit:

1. **Final report** (5-10 pages or equivalent) including:
    - Executive summary
    - Data preparation documentation
    - EDA findings
    - Visualization story
    - Interactive dashboard description
    - Bias and ethical considerations
    - Conclusions
    - **Optional ML extension** (if attempted)
2. **Final presentation** (8 minutes) showing:
    - What you did
    - What you found
    - What you recommend
    - What you learned

Example final report structure:

```
text
FINAL PROJECT REPORT
Titanic Passenger Survival Analysis
Student: [Your Name]
Date: December 3, 2026

1. EXECUTIVE SUMMARY
   - Goal: Understand factors affecting Titanic survival
   - Dataset: 891 passengers, 12 variables
   - Key finding: Class, sex, and age were strongest predictors
```

2. DATA PREPARATION
   - Cleaned dataset (dropped Cabin, imputed Age and Embarked)
   - 891 rows, 11 columns after cleaning
   - No remaining missing values

3. EXPLORATORY DATA ANALYSIS
   - 38% overall survival rate
   - Women survived at 74% vs 19% for men
   - 1st class: 63% survival, 3rd class: 24%
   - Children (<18): 54% survival
   - Significant age difference between survivors and non-survivors ($p<0.001$)

4. VISUALIZATION STORY
   - 5 static visualizations showing:
     - Survival by class (bar)
     - Age distribution (histogram)
     - Fare by class (box plot)
     - Survival by sex and class (grouped bar)
     - Correlation matrix (heatmap)

5. INTERACTIVE DASHBOARD
   - Plotly scatter plot with hover information
   - Filtering by class and sex
   - HTML file included

6. BIAS & ETHICAL CONSIDERATIONS
   - Historical bias: Women and children prioritized
   - Missing data: 77% of cabin data missing
   - Limitations: 1912 data may not generalize
   - Fairness: Analyzed subgroup performance

7. CONCLUSIONS
   - Strongest predictors: Sex, Class, Age
   - "Women and children first" policy evident
   - Wealth (class) strongly influenced survival
   - Interactive tools help explore complex patterns

8. OPTIONAL ML EXTENSION
   - Used Random Forest to predict survival

```
- 83% accuracy, 79% precision, 82% recall
- Feature importance: Sex > Class > Age > Fare
- Model performs well across subgroups
```

CHECKLIST

Before submitting, you should verify:

Required for ALL students:

- □ Dataset chosen and documented (Milestone 0)
- □ Data cleaned with documented decisions (Milestone 1)
- □ At least one statistical test performed (Milestone 1)
- □ At least 5 static visualizations created (Milestone 2)
- □ Chart design justified (Milestone 2)
- □ Bias critique included (Milestone 2)
- □ Interactive dashboard created (Milestone 3)
- □ Peer feedback incorporated (Milestone 3)
- □ Final report submitted (Week 16)
- □ Final presentation delivered (Week 16)

Optional for bonus points:

- □ Machine Learning extension included
- □ Advanced visualizations (maps, animations)
- □ Complete web deployment
- □ Additional data sources integrated